Razvan Mihai Popescu

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Education

Delft University of Technology — Delft, NL
Ph.D. in Computer Science, AI-enabled SE — *Mar. 2025 - Present*
Supervisors: Maliheh Izadi, Arie van Deursen

Delft University of Technology — Delft, NL
Master of Computer Science, Artificial Intelligence — *Sept. 2023 - Feb. 2025*
Courses: Supercomputing for Big Data, Software Architecture, Information Retrieval, Artificial Intelligence Techniques, Conversational Agents, Analytics and Machine Learning for Software Engineering, Testing and Validation for AI-Intensive Systems, Artificial Intelligence for Software Testing and Reverse Engineering, Natural Language Processing, Release Engineering for Machine Learning Applications

Delft University of Technology — Delft, NL
Bachelor of Computer Science and Engineering — *Sept. 2020 - Jul. 2023*
Courses: Algorithms and Data Structures, Object-oriented Programming, Software Engineering Methods, Software Quality and Testing, Big Data Processing, Data Mining, Machine Learning, Computational Intelligence, Collaborative Artificial Intelligence, Human-Computer Interaction

Publications

SWE-Amend: Benchmarking Coding Agents on Real Code Review — *Under Review*
Nawmi Nujhat, ***Razvan Mihai Popescu****, Maliheh Izadi*

Merged, but Maintained? Refactoring and Code Quality in Agent-Authored Pull Requests — *Under Review*
Inaesh Joshi, ***Razvan Mihai Popescu****, Maliheh Izadi*

Characterizing Human-Agent Dynamics in Agent-authored Pull Requests — *Under Review*
Ziyou Li, ***Razvan Mihai Popescu****, Maliheh Izadi*

When the Docstring Lies, LLM Judges Blame the Code — *Under Review*
Razvan Mihai Popescu*, Maliheh Izadi*

Evaluating Large Language Models in Code-Related Tasks: A Systematic Literature Review — *Under Review*
Razvan Mihai Popescu*, Maliheh Izadi*

Reliable and Developer-Aligned Evaluation of Agents for Software Engineering — *FSE '26*
Razvan Mihai Popescu

The Poisoned Chalice of LLM Evaluation Report — *FSE '26*
Jonathan Katzy, Ali Al-Kaswan, ***Razvan Mihai Popescu****, Zhou Yang*

Investigating Autonomous Agent Contributions in the Wild: Activity Patterns and Code Change over Time — *MSR '26*
Razvan Mihai Popescu*, David Gros, Andrei Botocan, Rahul Pandita, Prem Devanbu, Maliheh Izadi*

Silent Reliance or Diligent Review? Human-Agent Engagement and Interaction Patterns in GitHub — *ICSE '26*
Ziyou Li, ***Razvan Mihai Popescu****, Maliheh Izadi*

Automated Attention Pattern Discovery at Scale in Large Language Models (extension) — *TMLR '26*
Jonathan Katzy, ***Razvan Mihai Popescu****, Erik Mekkes, Arie van Deursen, Maliheh Izadi*

Automated Attention Pattern Discovery at Scale — *NeurIPS '25*
Jonathan Katzy, ***Razvan Mihai Popescu****, Erik Mekkes, Arie van Deursen, Maliheh Izadi*

The Heap: A Contamination-Free Multilingual Code Dataset for Evaluating Large Language Models — *FORGE '25*
Jonathan Katzy, ***Razvan Mihai Popescu****, Arie van Deursen, Maliheh Izadi*

An Exploratory Investigation into Code License Infringements in LLM Training Datasets — *FORGE '24*
Jonathan Katzy, ***Razvan Mihai Popescu****, Arie van Deursen, Maliheh Izadi*

Language Models for Code Completion: A Practical Evaluation *ICSE '24*
Maliheh Izadi, Jonathan Katzy, Tim van Dam, Marc Otten, ***Razvan Mihai Popescu****, Arie van Deursen*

A Study on the Impact of Common Code Structures on CodeParrot's Autocompletion Performance *TU Delft*
Razvan Mihai Popescu

Research Experience

Doctoral Researcher Mar. 2025 – Present
AISE Lab *TU Delft, NL*

- Analyzed 5M agent and human contributions, finding that agents refactor less and exhibit more code smells
- Evaluated 50K verdicts from five LLM judges, showing false docstrings cut acceptance of correct code by up to 82%
- Tracked 110K agent and human contributions, revealing lower survival and higher churn for agent-authored code
- Identified limited human oversight of coding agents, with 74% of 212K agent pull requests merged without modification
- Proposed a doctoral research plan for developer-aligned, real-world evaluation of LLM-powered coding agents
- Conducted a systematic literature review on evaluation techniques and experimental settings for LLMs in SE

Research Assistant Mar. 2023 – Mar. 2025
AISE Lab *TU Delft, NL*

- Co-developed AP-MAE, a vision transformer for scalable LLM attention analysis, achieving a 13.6% accuracy gain
- Built The Heap, a contamination-free code corpus spanning 57 programming languages for fair LLM evaluation
- Audited public LLM training datasets for license compliance, revealing infringement rates of up to 22%
- Contributed to Code4Me, an open-source IDE extension for evaluating LLM code completion in real-world settings
- Showed that common code structures reduce CodeParrot's null attention by 30% and prediction depth by 50%

Talks & Presentations

- *Reliable and Developer-Aligned Evaluation of Agents for Software Engineering*, FSE '26, Montreal, Canada
- *Investigating Autonomous Agent Contributions in the Wild*, MSR '26, Rio de Janeiro, Brazil
- *The Heap: A Contamination-Free Multilingual Code Dataset*, FORGE '25, Ottawa, Canada

Academic Service

- Program committee member of the Mining Challenge at MSR '27
- Co-chair of the Membership Inference Attack Competition at FSE '26 and ICSE '27
- Web chair of the AgenticSE Workshop at ASE '25
- Reviewer for ICSE, FSE, ASE, ISSTA, and TOSEM

Teaching & Supervision

Thesis Supervisor Mar. 2025 – Present
AISE Lab *TU Delft, NL*

- Supervised Nawmi Nujhat's MSc thesis on benchmarking coding agents on real-world code reviews (*SWE-Amend*)
- Supervised Inaesh Joshi's MSc thesis on agentic refactoring and code quality with Meta (*Merged, but Maintained?*)
- Supervised 4 BSc students on how data smells such as boilerplate, non-English code, and technical debt affect LLMs

Teaching Assistant Feb. 2025 – Apr. 2025
Responsible Data Science and AI Engineering *TU Delft, NL*

- Facilitated in-class discussions on the ethical and societal implications of AI systems in a 400-student course
- Graded student reports and final projects on the design of responsible AI solutions
- Co-organized and assessed final project presentations, providing feedback to student teams

Teaching Assistant Nov. 2024 – Feb. 2025
Machine Learning for Software Engineering *TU Delft, NL*

- Designed research project proposals on LLM-driven software engineering
- Mentored 4 student teams through project implementation and academic paper writing
- Evaluated student papers and final presentations on technical implementation and research quality

Master Mentor Sep. 2024 – Jan. 2025
MSc Computer Science *TU Delft, NL*

- Guided 15 first-year students through their transition into the program during the Master Your Start event
- Held weekly meetings on specialization choice, study planning, research groups, and course selection
- Hosted social events to build community and connections among incoming students

Head Teaching Assistant Feb. 2023 – Apr. 2023, Sep. 2024 – Nov. 2024
Software Architecture *TU Delft, NL*

- Led the teaching assistant team for a 260-student course, coordinating student support
- Advised 10 student teams weekly on designing and documenting architectures for socially relevant systems
- Assessed essays, proof-of-concept implementations, and final project presentations
- Managed the development and communication infrastructure for all 65 project groups

Seminars & Summer Schools

- Dagstuhl Seminar on Evaluation of AI Models in Software Engineering '26, Schloss Dagstuhl, Germany
- Machine Learning Summer School '26, hosted by Columbia University and Bloomberg, New York, USA
- Machine Learning Summer School '26, hosted by Maincode, Melbourne, Australia
- SEN Symposium '25 and AIWare BootCamp '25, Amsterdam, The Netherlands

Industry Experience

Full-Stack Developer Oct. 2021 – Jul. 2023
Lunar Zebro *Delft, NL*

- Designed a ground segment web application for the MINAR IX mission with the University of Edinburgh
- Developed a rover swarm control application with real-time sensor analysis, mapping, and object detection
- Improved rover control efficiency by 80% through real-time sensor data and video streaming
- Configured an asynchronous message broker to coordinate four lunar rovers simultaneously
- Represented TU Delft at the International Astronautical Congress '22, Paris, as the only student rover team

Software Engineer Apr. 2022 – Jul. 2022
Feedback-Analytics *The Hague, NL*

- Developed a word processor with form logic support, reducing report creation time by 30%
- Integrated the editor with the company's survey platform to generate customized feedback reports
- Created an automated reporting pipeline that renders client reports as PDFs and dispatches them via email

Projects

CNN-Based URL Phishing Detection Apr. 2024 - Jun. 2024

- Built an end-to-end release engineering pipeline taking a phishing detection model from training to production
- Implemented data and model versioning, infrastructure provisioning, and containerized deployment
- Set up automated testing and monitoring to track model quality and performance over time

RAG for Complex QA Jan. 2024 - Apr. 2024

- Developed a RAG pipeline using Contriever for retrieval and a quantized Llama 3 model for generation
- Evaluated the pipeline on 2WikiMultiHopQA to study how context type affects multi-hop answer generation
- Compared negative and random contexts against relevant retrieved passages, measuring their effect on answer quality

Conversational Movie Agent Nov. 2023 - Jan. 2024

- Built a movie recommendation agent with conversational memory on the Furhat social robot platform
- Designed the agent's long- and short-term memory and integrated entity recognition and sentiment analysis
- Ran a 50-participant user study showing that memory improved user satisfaction and recommendation accuracy

Prompt Augmentation for Code Generation Sep. 2023 - Nov. 2023

- Investigated whether partial software artifacts from the development context can steer LLMs toward correct code
- Benchmarked WizardCoder and CodeGPT on MBPP, comparing augmentation strategies across model sizes
- Raised pass@1 by 43% through test case augmentation, with the largest gains observed in bigger models

Technical Skills & Languages

Programming Languages: Python, Java, C++, JavaScript, Scala, SQL

Technologies & Frameworks: Git, Docker, AWS, Hugging Face, PyTorch, Scikit-learn, Apache Spark, Apache Kafka

Languages Spoken: Romanian (Native), English (C2), German (A2)